
Privacy Analysis of Augmented Reality Glasses

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Weekly Report 9

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What has been done (Previous Week)

- Finished the documentation for the set up for the static and dynamic analysis and uploaded
- Wrote documentation for the static analysis and uploaded, only missing the section 'API Usage'.
- Wrote documentation regarding the Bluetooth experiments though it is very basic, only including the meta data and an introduction for each experiment.
- Uploaded all the Bluetooth experiments and the voice recordings that were missing.
- Tried to use mitmproxy for analyzing network traffic with the work flow as so:
 - Start mitmproxy
 - Set the proxy on the phone
 - Installed and set up the certificate for mitmproxy on the phone.
 - Started interacting between the phone and glasses to generate traffic. I was able to see the traffic when I access google for example.
 - Problem: the Meta View app uses certificate pinning, and so the certificate is not accepted. This happens when I try to access spotify or ask the weather. I am still able to upload photos onto the phone.
 - The workarounds I tried are:
 - * 'apk-mitm' which tries to modify the app's source code, but the decoding of the APK failed.
 - * 'objection' which tries to bypass certificate pinning, but there were permissions issues.
 - * 'android unpinner' which tries to remove certificate pinning from the APK.
- Continued to try and decrypt Bluetooth traffic without rooting the phone. The simplified problems encountered is as so:
 - Tried to get the Bluetooth snopp log. It claimed to have no such file in the default '/sdcard/' location.
 - Then tried from other possible locations as it is not always in the exact same place, of which then I got a permissions denied.
 - Next, tried to configure the Android to log to a public directory so I can access it, but that failed due to permissions. The reason is because adb does not have root permissions on this phone.
 - So I tried to run-as to try to gain access to the specific directory that I want, but that was also denied due to permissions.

What will be done next week (and following weeks as some tasks may extend over multiple weeks)

- Try to generate a bug report for Bluetooth traffic for this phone. I was able to do this with the old phone and was able to open the bug report however when I apply any btle filter there was no packet, only the filter bluetooth had packets. Meaning that the capture pretty much only contained classic bluetooth.
- Continue to try and find workarounds for mitmproxy whilst not having the rooted phone.
- Continue to root the phone (once received the sim card) and if so write documentation about the process.
- Run a dynamic analysis again and write documentation if anything interesting happens.
- Document any result or analysis that can be done on the encrypted traffic from the experiments done in regards to the thesis goals.

Meeting

Meeting canceled.